

The NEW ENGLAND JOURNAL of MEDICINE

ESTABLISHED IN 1812

JULY 2, 2026

VOL. 395 NO. 1

Angiography-Derived Fractional Flow Reserve to Guide PCI

W.F. Fearon,^{1,2} A. Jeremias,^{3,4} G. Witberg,^{5,6} R. Al-Lamee,^{7,8} D.J. Cohen,^{3,4} A. Kaki,⁹ R.P. Sharma,¹ R.W. Yeh,¹⁰ B.M. Chehab,¹¹ M.C. Kim,¹² H. Otake,¹³ R. Tayal,¹⁴ H. Matsuo,¹⁵ M. McEntegart,¹⁶ A.K. Patel,¹⁷ Y. Sandoval,¹⁸ K.M. Al-Azizi,¹⁹ K. Dan,²⁰ L. Razzouk,²¹ S. Fournier,²² A. Popma,³ J.J. Popma,^{3,10} D. Zhao,³ B. Redfors,^{3,23-25} M.V. Madhavan,^{3,16} R. Kornowski,^{5,6,26} M.B. Leon,^{3,16} and A.J. Kirtane,^{3,16} for the ALL-RISE Investigators*

ABSTRACT

BACKGROUND

Assessing intermediate coronary lesions with an intracoronary pressure wire improves clinical outcomes in patients undergoing cardiac catheterization and percutaneous coronary intervention (PCI). However, clinical use of pressure-wire–based physiological assessment remains low. Measurement of fractional flow reserve (FFR) derived from coronary angiographic images alone correlates well with pressure-wire–based FFR measurements and may simplify procedures, but its effect on clinical outcomes is unknown.

METHODS

In this international noninferiority trial, we randomly assigned patients undergoing coronary angiography who were found to have at least one intermediate coronary stenosis to physiological assessment with measurements derived from angiographic images (FFRangio) or with pressure-wire–based measurements. The primary end point was a composite of death, myocardial infarction, or unplanned, clinically indicated coronary revascularization at 1 year. The noninferiority margin was 3.5 percentage points.

RESULTS

A total of 1930 patients were randomly assigned to physiological assessment with FFRangio (FFRangio group; 965 patients) or a pressure-wire–based approach (pressure-wire group; 965 patients). The mean age of the patients was 68.4 years, and 25.0% of the patients were women. At 1 year, a primary end-point event had occurred in 64 patients (Kaplan–Meier estimate, 6.9%) in the FFRangio group and 65 patients (Kaplan–Meier estimate, 7.1%) in the pressure-wire group (hazard ratio, 0.98; 95% confidence interval, 0.70 to 1.39; difference, –0.2 percentage points; upper boundary of the one-sided 97.5% confidence interval, 2.1 percentage points; $P < 0.001$ for noninferiority). There were no apparent differences between the groups with respect to the incidence of bleeding, acute kidney injury, or procedure-related adverse events.

CONCLUSIONS

Among patients with intermediate coronary-artery lesions undergoing physiological assessment in the cardiac catheterization laboratory, an angiography-guided strategy involving FFRangio was noninferior to a pressure-wire–guided strategy with respect to a composite end point of death, myocardial infarction, or unplanned clinically indicated coronary revascularization at 1 year. (Funded by CathWorks; ALL-RISE ClinicalTrials.gov number, NCT05893498.)

The authors' full names, academic degrees, and affiliations are listed at the end of the article. William F. Fearon can be contacted at wfearon@stanford.edu or at Stanford University School of Medicine, 300 Pasteur Dr., H2103, Stanford, CA 94305.

*A list of the ALL-RISE Trial Investigators is provided in the Supplementary Appendix, available at NEJM.org.

William F. Fearon and Allen Jeremias contributed equally to this article.

This article was published on March 29, 2026, at NEJM.org.

N Engl J Med 2026;395:9-19.

DOI: 10.1056/NEJMoa2600949

Copyright © 2026 Massachusetts Medical Society.

CME



 A Quick Take
is available at
NEJM.org



SEVERAL RANDOMIZED TRIALS HAVE shown the benefit of calculating fractional flow reserve (FFR; the ratio, at maximal hyperemia, of the mean arterial blood pressure measured distal to a stenosis to the mean pressure measured before the stenosis) with a coronary-artery pressure wire to evaluate the physiological importance of coronary-artery lesions and guide revascularization decisions in patients with coronary artery disease.¹⁻³ Subsequent studies have shown that the use of a nonhyperemic pressure ratio (also known as instantaneous wave-free ratio) to assess lesions can reduce the need for a pharmacologic agent to induce maximal hyperemia during testing and results in outcomes that are similar to those obtained with FFR measurement.^{4,5} These data have led to class I recommendations in multiple society guidelines for the use of FFR or a nonhyperemic pressure ratio in the assessment of intermediate coronary stenoses found on coronary angiography.^{6,7} However, incorporation of pressure-wire–based physiological assessment of coronary-artery lesions during cardiac catheterization procedures remains lower than expected (<20% of cases), owing in part to workflow considerations. Pressure-wire–based assessment requires a coronary guide catheter for delivery of a dedicated intracoronary pressure-wire system that is advanced distal to the stenosis of interest and in some cases involves administration of a pharmacologic agent to induce maximal hyperemia.⁸

Computer-based software (CathWorks) has been developed to facilitate the assessment of coronary physiological features in the cardiac catheterization laboratory without the use of an intracoronary pressure wire. This computer-based software analyzes coronary angiographic images to create a three-dimensional reconstruction of the coronary-artery tree and apply a resistance model to estimate pressure gradients on the basis of vessel contours. The system (FFRangio) quantifies angiography-derived FFR, which correlates well with pressure-wire–derived FFR but does not require a pressure wire or hyperemic agent.⁹⁻¹¹ We conducted the Advancing Cath Lab Results with FFRangio Coronary Physiology Assessment (ALLRISE) trial to determine whether clinical outcomes after physiological assessment of coronary stenoses with FFRangio were noninferior to outcomes after pressure-wire–based assessment to guide revascularization decision making.

METHODS

TRIAL DESIGN

We conducted this international, randomized, noninferiority trial at 59 sites in North America, Europe, Asia, and the Middle East. Details of the design and conduct of the trial have been published previously,¹² and the protocol is available with the full text of this article at NEJM.org. Clinical data and any adverse events were entered into an electronic case report form, and supporting documentation was provided. All adverse events were adjudicated by an independent clinical-events committee whose members were unaware of the group assignments. An independent data and safety monitoring committee oversaw the conduct of the trial. The trial was sponsored by CathWorks, which participated in site selection, data collection, and monitoring. Statistical analysis was performed independently by the Cardiovascular Research Foundation. The principal investigators and steering committee monitored all aspects of the trial. The principal investigators and trial chairperson had unrestricted access to the data, prepared all drafts of the manuscript before submitting it for publication, and vouch for the completeness and accuracy of the data and the fidelity of the trial to the protocol. The trial protocol was approved by the relevant institutional review boards or ethics committees, and all patients provided written informed consent before undergoing diagnostic coronary angiography.

PATIENTS

Patients who were at least 18 years of age and were undergoing invasive coronary angiography were eligible if they met the main inclusion criterion of having, on the basis of visual assessment, at least one coronary stenosis of 50 to 90%, indicating a need for hemodynamic evaluation with FFR or a nonhyperemic pressure ratio to guide coronary revascularization. Key exclusion criteria were a recent ST-segment elevation myocardial infarction, culprit vessel in a patient with a non-ST-segment elevation acute coronary syndrome, candidacy for coronary-artery bypass graft surgery, previous coronary-artery bypass grafting that involved a bypass graft to the study vessel (the vessel containing the lesion designated for assessment), severe aortic or mitral valvular heart disease, or severe left ventricular dysfunction. A complete listing of the inclusion and exclusion

criteria is provided in Table S1 in the Supplementary Appendix, available online at NEJM.org.

RANDOMIZATION AND FOLLOW-UP

Eligible participants were randomly assigned in a 1:1 ratio to receive coronary-artery lesion assessment with the use of FFRangio (FFRangio group) or a coronary pressure wire (pressure-wire group). Before randomization, investigators identified which lesion or lesions would be the subject of physiological assessment (the study lesion) and successfully treated any nonstudy lesions that required percutaneous coronary intervention (PCI) without physiological assessment. Randomization was stratified on the basis of clinical presentation (stable angina or a non–ST-segment elevation acute coronary syndrome) and whether the investigator planned to measure FFR or a nonhyperemic pressure ratio if the patient were to be assigned to pressure-wire assessment. Patients were followed-up during their hospitalization and at 30 days, 6 months, and 1 year.

TRIAL PROCEDURES

Diagnostic coronary angiography was performed at each site. The trial design included a recommendation that three or four angiograms of each vessel be obtained, with images being at least 30 degrees apart, vessel overlap and foreshortening minimized, and the entire vessel imaged without movement of the catheterization-laboratory table.

Patients assigned to the pressure-wire group could undergo either FFR testing or nonhyperemic pressure ratio measurement with any commercially available pressure-wire system. If FFR was measured, either intracoronary or intravenous adenosine or adenosine triphosphate or intracoronary papaverine at recommended doses could be used to induce hyperemia. An FFR value of up to 0.80 or a nonhyperemic pressure ratio value of up to 0.89 was used to indicate a functionally significant lesion warranting PCI, whereas lesions with values above these cutoffs were to be treated medically.

For patients assigned to the FFRangio group, angiography-derived FFR was calculated by the trial investigators at each site. Three images of the coronary artery were selected, and the study lesion was identified in each image. The system automatically traced the lumen borders of each vessel, which was reviewed and manually corrected, if necessary. The system then processed the

data, created a three-dimensional reconstruction of the coronary-artery tree, and calculated the FFR of each vessel. For assessments in this group, a value of up to 0.80 was used to indicate a functionally significant lesion warranting PCI, whereas lesions with a value above this cutoff were to be treated medically.

PCI was performed according to the standard of care at each site. Intravascular imaging was encouraged, but not required, to assist with potential improvement of the PCI result. Coronary physiological assessment with either trial strategy after PCI was not permitted for guiding PCI. Coronary angiograms and physiological assessments were sent digitally to an independent core laboratory for review and analysis.

END POINTS

The primary end point was a composite of death from any cause, myocardial infarction, or unplanned, clinically indicated revascularization at 1 year. Key secondary end points were the outcome of each component of the primary end point, procedural time, radiation exposure, and use of contrast media. A list of primary and secondary end points is provided in Table S2, and end-point definitions are provided in Table S3 and the protocol.

STATISTICAL ANALYSIS

The primary analysis was conducted in accordance with the intention-to-treat principle. The 1-year event estimate for the primary end point was 7.5% in both groups.^{4,5} On the basis of previous studies that used a similar margin and the steering committee's assessment of a clinically acceptable difference given the inclusion of repeat revascularization in the primary end point, an absolute noninferiority margin of 3.5 percentage points was specified.^{4,5,13} We estimated that a sample size of 1924 patients would provide the trial with more than 80% power to show noninferiority of the FFRangio strategy, at a one-sided P value of less than 0.025, assuming a 5% loss to follow-up.

The primary end point was analyzed with Kaplan–Meier methods to estimate the 1-year cumulative incidence of a primary end-point event. Confidence intervals for differences in the Kaplan–Meier estimates were obtained with Greenwood's formula. The Com-Nougue method was applied to test the one-sided noninferiority hypothesis by calculating whether the difference in event

probabilities remained within the prespecified noninferiority margin. For the analysis of the primary end point, no imputation for missing data was performed. Data from patients who did not complete the 12-month follow up and remained event-free were censored at the date of the patients' last known status. Competing risk analyses were performed with Fine–Gray subdistribution hazard models, with death as a competing event. No interim analyses were planned.

Because the trial was powered only for the primary end point, the overall type I error was controlled, and no adjustment for multiplicity was required. Exploratory subgroup analyses of the primary end point used Cox proportional-hazards models. Potential heterogeneity in the treatment effect across sites and geographic regions was assessed by fitting Cox proportional-hazards models that included treatment group, site (or region), and the treatment-by-site (or treatment-by-region) interaction term as fixed effects. Analyses were performed with SAS statistical software, version 9.4 or later (SAS Institute). Additional details are provided in the protocol (which includes the statistical analysis plan) and the Supplementary Appendix.

RESULTS

PATIENTS AND ASSESSMENTS

From June 21, 2023 to January 2, 2025, a total of 4704 patients were screened for eligibility, and 1930 patients were randomly assigned to FFRangio (965 patients) or pressure-wire–based (965 patients) assessment (Fig. S1 and Table S4). The clinical characteristics of the patients at baseline are shown in Table 1 and appeared to be well matched between the two groups. The mean ($\pm$ SD) age of the participants was 68.4 ± 10.2 years, and 25.0% were women. Most of the patients (90.6%) presented without elevated cardiac biomarkers, and 37.8% of the patients had a history of diabetes. Overall, the demographic and clinical characteristics of the enrolled population were representative of the broader population of persons with coronary artery disease and those undergoing invasive testing in the geographic regions in which the trial was conducted (Table S5).

The baseline coronary angiographic and physiological characteristics are shown in Table 2.

Most of the patients had a single lesion for assessment; the left anterior descending artery was identified as the vessel containing the study lesion in 49.6% of patients. Assessment with FFRangio was conducted successfully in 98.5% of lesions, with a mean value of 0.81 ± 0.12 , and 43.0% of measurements had an abnormal value (≤ 0.80). Pressure-wire–based physiological measurements were made successfully in 98.6% of lesions; FFR assessment was performed in 55.5% of those lesions, and nonhyperemic pressure ratio assessment alone was performed in 44.5%. The mean FFR value was 0.84 ± 0.10 and the mean nonhyperemic pressure ratio was 0.89 ± 0.11 , with 37.0% of the lesions that were measured with a pressure wire having an abnormal value (≤ 0.80 for FFR and ≤ 0.89 for nonhyperemic pressure ratio). Distributions of the measurement values are shown in Figure S2.

The median time to calculate the assessment was 6 minutes (interquartile range, 4 to 10) in the FFRangio group, as compared with 8 minutes (interquartile range, 5 to 15) in the pressure-wire group; the between-group difference was -2 minutes (95% confidence interval [CI], -3 to -2). PCI was performed in 39.9% of all lesions. However, PCI was performed in 44.3% of lesions assessed with FFRangio and 35.4% assessed with the use of a pressure wire (odds ratio, 1.45; 95% CI, 1.24 to 1.70). The median overall procedural time in the FFRangio group was 39 minutes (interquartile range, 21 to 62) and in the pressure-wire group was 42 minutes (interquartile range, 29 to 60); the between-group difference was -5 minutes (95% CI, -7 to -3). The median total fluoroscopy time was 9 minutes (interquartile range, 5 to 17) in the FFRangio group and 12 minutes (interquartile range, 8 to 20) in the pressure-wire group; the median volume of contrast media used was 100 ml (interquartile range, 52 to 145) in the FFRangio group and 105 ml (interquartile range, 72 to 150) in the pressure-wire group. The between-group difference in fluoroscopy time was -3 minutes (95% CI, -3 to -2) and the between-group difference in contrast media volume was -14 ml (95% CI, -20 to -10).

END POINTS

Complete 1-year follow-up was obtained for 1853 of 1930 patients (96.0%) — 921 patients (95.4%) in the FFRangio group and 932 patients (96.6%)

Table 1. Characteristics of the Patients at Baseline.*

Characteristic	FFRangio (N=965)	Pressure Wire† (N=965)
Age — yr	68.4±10.1	68.4±10.3
Sex — no. (%)		
Male	729 (75.5)	719 (74.5)
Female	236 (24.5)	246 (25.5)
Race or ethnic group — no. (%)‡		
White	576 (59.7)	596 (61.8)
Asian	238 (24.7)	227 (23.5)
Black	54 (5.6)	49 (5.1)
American Indian or Alaska Native	2 (0.2)	2 (0.2)
Native Hawaiian or Pacific Islander	1 (0.1)	3 (0.3)
Other	35 (3.6)	34 (3.5)
Not reported	59 (6.1)	54 (5.6)
Body-mass index§	28.2±5.8	28.2±5.7
Diabetes — no./total no. (%)	381/964 (39.5)	349/962 (36.3)
Insulin dependent — no./total no. (%)	130/964 (13.5)	111/962 (11.5)
Hypertension — no./total no. (%)	757/964 (78.5)	769/962 (79.9)
Dyslipidemia — no./total no. (%)	759/964 (78.7)	764/962 (79.4)
Smoking status — no./total no. (%)		
Current tobacco user	150/964 (15.6)	124/962 (12.9)
Previous tobacco user	343/964 (35.6)	377/962 (39.2)
Family history of CAD — no./total no. (%)	324/964 (33.6)	371/962 (38.6)
Previous myocardial infarction — no./total no. (%)	166/964 (17.2)	158/962 (16.4)
Previous PCI — no./total no. (%)	375/964 (38.9)	375/962 (39.0)
Renal disease — no./total no. (%)	186/944 (19.7)	169/944 (17.9)
Ejection fraction ≤50% — no./total no. (%)	109/799 (13.6)	120/822 (14.6)
Hospitalized with NSTEMI-ACS — no. (%)	96 (9.9)	85 (8.8)

* Plus-minus values are means ±SD. CAD denotes coronary artery disease, FFR fractional flow reserve, FFRangio angiography-derived FFR, IQR interquartile range, NSTEMI-ACS non-ST-segment elevation acute coronary syndrome, and PCI percutaneous coronary intervention.

† Wire-based assessment included nonhyperemic pressure ratio measurement with any commercially available pressure-wire system.

‡ Race and ethnic group were reported by the patient.

§ The body-mass index is the weight in kilograms divided by the square of the height in meters.

in the pressure-wire group. At 1 year, a primary end-point event had occurred in 64 patients (Kaplan–Meier estimate, 6.9%) in the FFRangio group and 65 patients (Kaplan–Meier estimate, 7.1%) in the pressure-wire group (hazard ratio, 0.98; 95% CI, 0.70 to 1.39; difference, −0.2 percentage points; upper boundary of the one-sided 97.5% confidence interval, 2.1 percentage points; $P<0.001$ for noninferiority) (Fig. 1A and Fig. S3).

There was no apparent evidence of heterogeneity of treatment effects across geographic regions (Table S6). The incidence of each of the individual components of the primary end point are shown in Table 3 and Figure 1. The results of the competing risk analysis were similar to those from the cause-specific Cox proportional-hazards models and supported the same conclusions (Table S7). The results appeared to be consistent

Table 2. Procedural and Angiographic Characteristics.*

Characteristic	FFRangio (N=965)	Pressure Wire (N=965)
Patients with nonstudy lesions receiving PCI — no. (%)	166 (17.2)	138 (14.3)
Median contrast media amount used (IQR) — ml	100 (52–145)	105 (72–150)
Median fluoroscopy time (IQR) — min	9 (5–17)	12 (8–20)
Median procedure duration (IQR) — min	39 (21–62)	42 (29–60)
Median physiological assessment duration (IQR) — min	6 (4–10)	8 (5–15)
Study lesions — no.	1288	1248
Location — no./total no. (%)		
Left anterior descending	635/1288 (49.3)	624/1248 (50.0)
Left circumflex	277/1288 (21.5)	261/1248 (20.9)
Right coronary	376/1288 (29.2)	363/1248 (29.1)
Reference vessel diameter — mm	2.9±0.6	2.9±0.6
Preprocedure diameter stenosis — %	54.5±12.0	54.5±12.0
Lesion length (IQR) — mm	12.0 (8.0–19.1)	12.5 (8.0–18.8)
Successful physiological assessment — no. (%)	1269/1288 (98.5)	1230/1248 (98.6)
FFRangio-based FFR	0.81±0.12	NA
Wire-based FFR	NA	0.84±0.10
Wire-based NHPR	NA	0.89±0.11
FFRangio-based FFR ≤0.80 — no./total no. (%)	545/1268 (43.0)	NA
Wire-based FFR ≤0.80 or NHPR ≤0.89 — no. (%)	NA	454/1228 (37.0)
No. of study lesions treated with PCI — no. (%)	571/1288 (44.3)	442/1248 (35.4)
Intravascular imaging use — no./total no. (%)	280/571 (49.0)	231/442 (52.3)

* Plus-minus values are means ±SD. Lesions that would be assessed (study lesions) and those that would not be assessed (nonstudy lesions) were designated by the investigators at the start of the trial. NA denotes not applicable, and NHPR nonhyperemic pressure ratio.

across most of the prespecified subgroups with the possible exception of the subgroup with study lesions in the left anterior descending coronary artery and the subgroup comprising patients in whom investigators intended to use FFR or a non-hyperemic pressure ratio for the pressure-wire-based assessment (Fig. 2).

Safety data were similar in the two groups and are shown in Table 3. A Bleeding Academic Research Consortium (BARC) stage 3, 4, or 5 bleeding event (range, 0 [no bleeding] to 5 [fatal bleeding]) occurred in 7 patients (0.7%) in the FFRangio group and 14 (1.5%) in the pressure-wire group (hazard ratio, 0.50; 95% CI, 0.20 to 1.23). No patients in the FFRangio group and one patient in the pressure-wire group had acute kidney injury. Additional adverse events reported as of day 30 are shown in Table S8. Periprocedural angiographic

complications occurred in 10 patients (1.0%) in the FFRangio group and 5 patients (0.5%) in the pressure-wire group (hazard ratio, 2.00; 95% CI, 0.68 to 5.85).

DISCUSSION

This multicenter, international, randomized trial showed that using angiographic images and FFRangio alone to guide decision making regarding PCI was noninferior to the reference standard pressure-wire-based testing with respect to a composite end point of death from any cause, myocardial infarction, or unplanned clinically indicated revascularization at 1 year. Event rates for each component of the primary end point, including death and myocardial infarction, appeared to be similar between the two approaches.

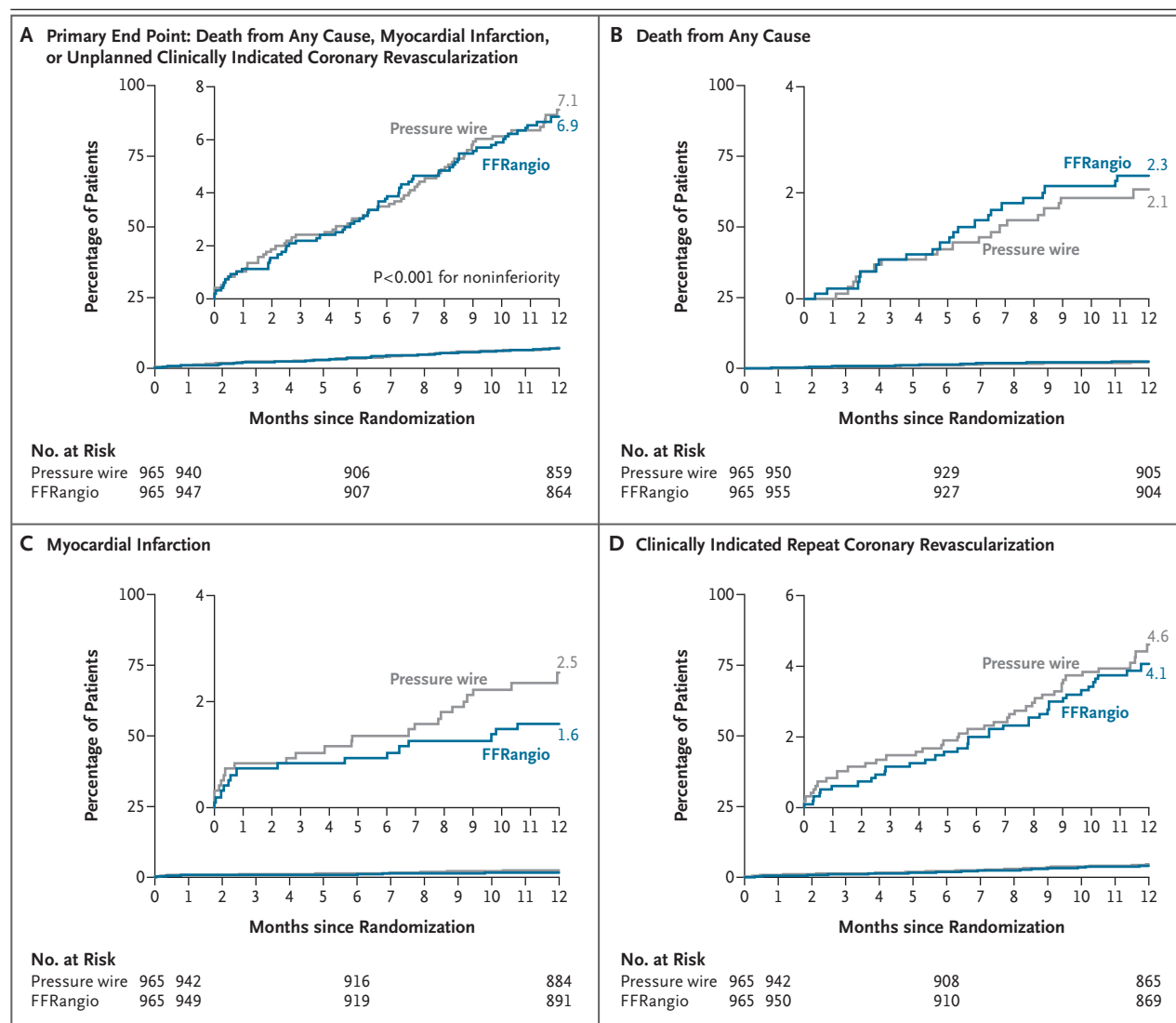


Figure 1. Composite Primary End Point and Individual Components at 12 Months (Kaplan–Meier Analysis).

Shown is the incidence, at 12 months after randomization, of the composite end point of death from any cause, myocardial infarction, or unplanned clinically indicated coronary revascularization (Panel A), and the individual components of death from any cause (Panel B), myocardial infarction (Panel C), and clinically indicated repeat coronary revascularization (Panel D). The two trial groups were patients who had angiography-derived assessment of fractional flow reserve (FFR; FFRangio group) and patients who had pressure-wire–based FFR or nonhyperemic pressure ratio measurement (pressure-wire group). The 12-month number at risk includes all patients for whom data were available at the 12-month follow-up (days 335 to 395). The insets in each panel show the same data on expanded y axes.

The FFRangio strategy reduced procedural time, radiation exposure, and use of contrast medium as compared with the pressure-wire–based assessment. However, there were more abnormal values and subsequent PCI procedures with the FFRangio approach than with the pressure-wire–based assessment. The higher rate of PCI in the FFRangio group might have been the result of lesions in which the measured value was near

the threshold for revascularization — a context in which studies have shown similar outcomes whether PCI is performed or deferred.¹⁴

The use of invasive coronary physiological assessment to guide revascularization decision making has been shown to improve outcomes as compared with conventional angiographic assessment.^{1,2} Assessment of coronary physiological features by software-based computation of

Table 3. End-Point Events at 1 Year.*

End-Point Event	FFRangio (N=965)	Pressure Wire (N=965)	Hazard Ratio (95% CI)
	<i>no. (%)</i>		
Composite primary end point: death, myocardial infarction, or unplanned clinically indicated revascularization†	64 (6.9)	65 (7.1)	0.98 (0.70–1.39)
Secondary end points‡			
Death from any cause	22 (2.3)	19 (2.1)	1.16 (0.63–2.14)
Death from cardiac causes	9 (1.0)	7 (0.7)	1.29 (0.48–3.46)
Myocardial infarction	15 (1.6)	23 (2.5)	0.65 (0.34–1.25)
Spontaneous	12 (1.3)	20 (2.2)	0.60 (0.29–1.22)
Procedural	3 (0.3)	3 (0.3)	1.00 (0.20–4.94)
Unplanned clinically indicated revascularization	37 (4.1)	41 (4.6)	0.90 (0.58–1.40)
PCI	32 (3.5)	39 (4.4)	0.82 (0.51–1.30)
Coronary bypass surgery	5 (0.6)	4 (0.5)	1.26 (0.34–4.68)
Definite or probable stent thrombosis	2 (0.2)	4 (0.5)	0.50 (0.09–2.72)
Safety end points			
BARC stage 3, 4, or 5 bleeding§	7 (0.7)	14 (1.5)	0.50 (0.20–1.23)
Acute kidney injury	0	1 (0.1)	NA

* Shown are the number of patients with events; the percentages (with the exception of death for the secondary end point) are Kaplan–Meier estimates at 1 year.

† $P=0.0008$ for noninferiority is based on the comparison of the upper boundary of a one-sided 97.5% confidence interval of risk difference against a threshold of 3.5 percentage points (difference, -0.2 percentage points; upper boundary of the one-sided 97.5% confidence interval, 2.1 percentage points).

‡ For secondary end points, the widths of the confidence intervals have not been adjusted for multiplicity and should not be used to infer treatment effect.

§ Bleeding Academic Research Consortium (BARC) staging of bleeding events ranges from 0 (no bleeding) to 5 (fatal bleeding).

physiological measurements derived from angiographic images instead of a pressure wire could potentially improve intraprocedural workflow, which could increase the adoption of physiological assessment. Deferring revascularization on the basis of a negative FFRangio value has been shown to result in a favorable clinical outcome,¹⁵ and a randomized trial in Japan showed that FFRangio-derived measurements resulted in a percentage of patients receiving PCI that was similar to that with FFR assessment, with similar clinical outcomes at 1 year.¹⁶ However, larger-scale prospective comparative data have been lacking. The finding of similar clinical outcomes with FFRangio assessment and pressure-wire–based assessment supports the use of this technology as a guide to revascularization decision making in appropriate patients.

The results of our trial differ from those in

the FAVOR (Functional Diagnostic Accuracy of Quantitative Flow Ratio in Online Assessment of Coronary Stenosis) III Europe trial, a multicenter, randomized trial that compared a coronary angiography-derived physiological assessment system to a pressure-wire–based strategy.¹³ In the FAVOR III Europe trial, measurement of the quantitative flow ratio to guide coronary revascularization was not noninferior to a pressure-wire–based FFR strategy, with worse clinical outcomes in both the patients who received PCI on the basis of the quantitative flow ratio result and those who had deferral of revascularization.¹⁷ The explanation for the discordant results between these two studies may be the different methods used for FFRangio assessment and for calculation of quantitative flow ratio as well as the fact that calculation of the quantitative flow ratio may require more operator interaction with

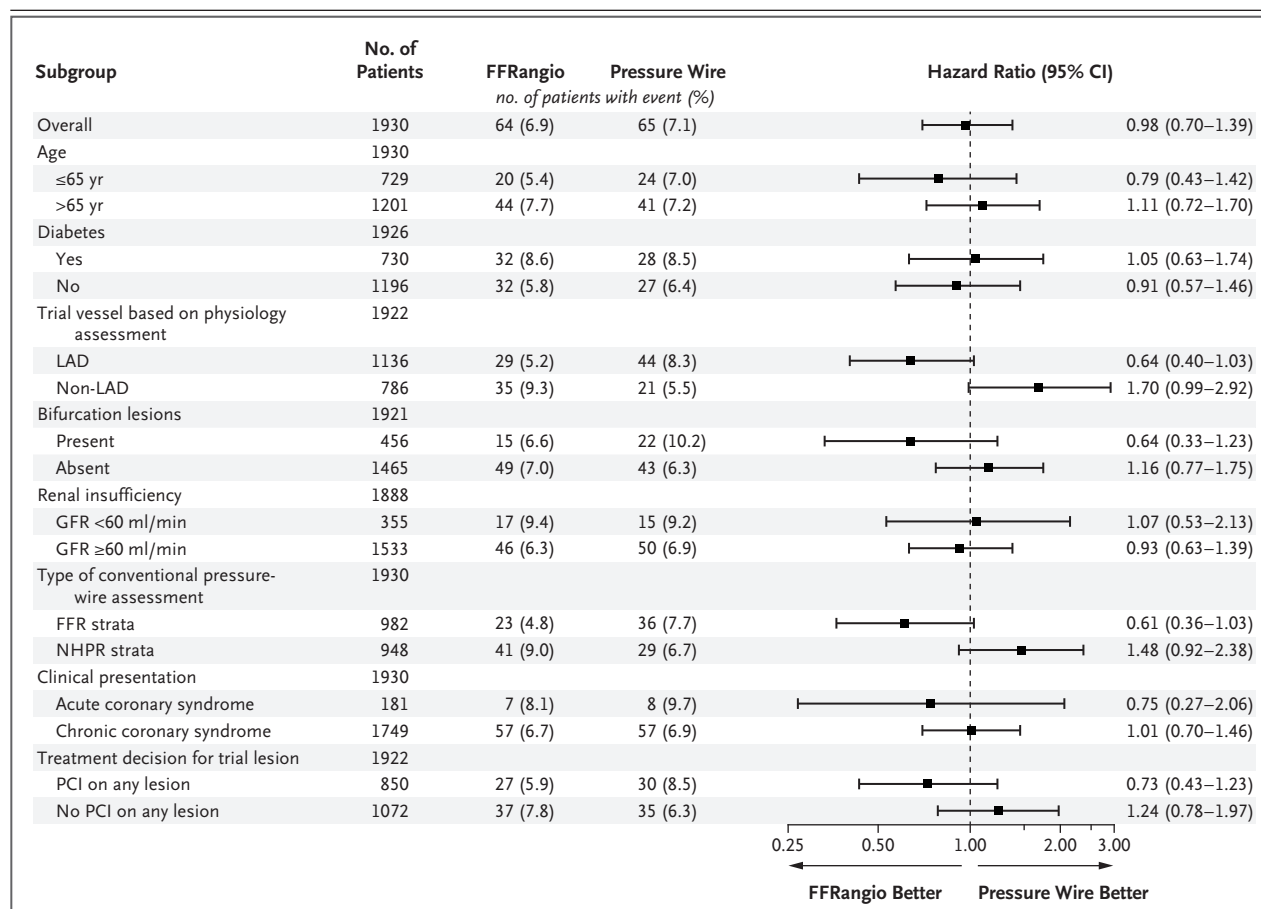


Figure 2. Subgroup Analyses of the Primary End Point.

Shown are results of subgroup analyses of the composite primary end point made with the use of Cox proportional-hazards models. The widths of the confidence intervals have not been adjusted for multiplicity and should not be used to infer treatment effect. GFR denotes glomerular filtration rate, LAD left anterior descending, NHPR nonhyperemic pressure ratio, and PCI percutaneous coronary intervention.

the software and manual adjustments, which could result in interobserver variability. For example, a substudy in the FAVOR III Europe trial suggested a high degree of variability in the calculation of the quantitative flow ratio between the participating sites and the core laboratory,¹⁸ whereas a strong correlation has been shown between FFRangio results calculated by a core laboratory and those calculated at multiple other sites.¹⁰ In a design similar to that of our trial, the Fractional Flow Reserve or 3D-Quantitative-Coronary-Angiography–Based Vessel-FFR–Guided Revascularization (FAST III) trial is comparing another method for calculating the angiography-derived vessel FFR with pressure-wire–derived FFR and will provide further data on this topic.¹⁹

Although the overall trial findings appear to be consistent across a variety of subgroups, subgroup analyses indicate the possibility of a differential effect when results in the left anterior descending artery are compared with results in other coronary vessels; the incidence of events was lower with FFRangio assessment than with pressure-wire–based assessment when applied to lesions in the left anterior descending artery and was higher in lesions in other vessels. Our findings also suggest a possible difference in outcome depending on whether the pressure-wire–based comparator strategy was based on FFR or nonhyperemic pressure ratio. This interaction may have been driven by random differences in absolute event rates in the two FFRangio-guided

strata; however, our trial was underpowered for such an assessment.

We noted a greater incidence of PCI among patients in the FFRangio group than among those in the pressure-wire group. It is important to consider that investigators specified which lesion or lesions would have physiological assessment before randomization. The higher rate of PCI in the FFRangio group may have led to PCI in fewer lesions during the follow-up period, although additional treatment could also have increased the chance of restenosis leading to repeat revascularization.

Limitations of our trial include the fact that although the inclusion criteria were broad, the investigators had to first determine that the coronary anatomy was amenable to both pressure-wire–based and FFRangio-based measurements. The majority of patients included in the trial had stable coronary disease or no biomarker elevation. The role of this strategy in assessment of nonculprit lesions in patients with non–ST-segment elevation acute coronary syndromes and ST-segment elevation myocardial infarction is being tested in the AIR-STEMI trial (Functional Coronary Angiography to Indicate and Guide Revascularization in STEMI Patients with Multivessel Disease).²⁰ The comparison of a FFRangio-based strategy to a mixed comparison of both pressure-wire–derived FFR and nonhyperemic pressure ratio could be considered a limitation because there are more data supporting the use of FFR than nonhyperemic pressure ratio. However, our trial is representative of current clinical use of FFR and nonhyperemic pressure ratio, and in designing the trial we aimed to model real-world practice. In addition, trial-group assignment was unblinded, which would be expected to favor the established standard of care in trials testing a new diagnostic approach. The very similar rates of unplanned clinically indicated revascularization in the two groups are therefore reassuring in this respect. The event rate for the primary end point in the pressure-wire (control) group was 7.1%, which is lower than the Kaplan–Meier estimate of 7.5%, and the actual achieved power of the trial was 87.3%. In addition, the upper bound of the 97.5% confidence interval for the primary end point was 2.1 percentage points, which is below the noninferiority margin of 3.5 percentage points.

Among patients undergoing physiological assessment to guide coronary revascularization, an FFRangio-guided strategy for assessing whether PCI should be performed in patients with intermediate coronary disease was noninferior to the reference standard of coronary pressure-wire–guided assessment with respect to clinical outcomes at 1 year.

Supported by CathWorks.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

A data sharing statement provided by the authors is available with the full text of this article at NEJM.org.

AUTHOR INFORMATION

William F. Fearon, M.D.,^{1,2} Allen Jeremias, M.D.,^{3,4} Guy Witberg, M.D., M.P.H.,^{5,6} Rasha Al-Lamee, M.D., Ph.D.,^{7,8} David J. Cohen, M.D.,^{3,4} Amir Kaki, M.D.,⁹ Rahul P. Sharma, M.D., M.B., B.S.,¹ Robert W. Yeh, M.D.,¹⁰ Bassem M. Chehab, M.D.,¹¹ Michael C. Kim, M.D.,¹² Hiromasa Otake, M.D.,¹³ Rajiv Tayal, M.D.,¹⁴ Hitoshi Matsuo, M.D., Ph.D.,¹⁵ Margaret McEntegart, M.D.,¹⁶ Ankitkumar K. Patel, M.D., M.P.H.,¹⁷ Yader Sandoval, M.D.,¹⁸ Karim M. Al-Azizi, M.D.,¹⁹ Kazuhiro Dan, M.D., Ph.D.,²⁰ Louai Razzouk, M.D., M.P.H.,²¹ Stephane Fournier, M.D.,²² Alexandra Popma, M.D.,³ Jeffrey J. Popma, M.D.,^{3,10} Duzhi Zhao, M.S.,³ Bjorn Redfors, M.D., Ph.D.,^{3,23-25} Mahesh V. Madhavan, M.D.,^{3,16} Ran Kornowski, M.D.,^{5,6,26} Martin B. Leon, M.D.,^{3,16} and Ajay J. Kirtane, M.D.^{3,16}

¹Division of Cardiovascular Medicine and Stanford Cardiovascular Institute, Stanford University, Stanford, CA; ²Veterans Affairs Palo Alto Health Care System, Palo Alto, CA; ³Cardiovascular Research Foundation, New York; ⁴Department of Cardiology, St. Francis Hospital and Heart Center, Roslyn, NY; ⁵Department of Cardiology, Rabin Medical Center, Petah Tikva, Israel; ⁶Gray School of Medicine, Tel Aviv University, Tel Aviv, Israel; ⁷National Heart and Lung Institute, Imperial College London, London; ⁸Imperial College Healthcare NHS Trust, London; ⁹Interventional Cardiology Department, Ascension St. John Hospital and Center, Detroit; ¹⁰Division of Cardiovascular Medicine, Smith Center for Outcomes Research in Cardiology, Beth Israel Deaconess Medical Center, Harvard Medical School, Boston; ¹¹Ascension Via Christi Hospital, University of Kansas, Wichita; ¹²Northwell Health, Lenox Hill Hospital, New York; ¹³Division of Cardiovascular Medicine, Department of Internal Medicine, Kobe University Graduate School of Medicine, Kobe, Japan; ¹⁴Valley Health Hospital, Paramus, NJ; ¹⁵Department of Cardiovascular Medicine, Gifu Heart Center, Gifu, Japan; ¹⁶Columbia University Irving Medical Center of the NewYork–Presbyterian Hospital, New York; ¹⁷Hackensack Meridian School of Medicine, Hackensack Meridian Mountaintop Medical Center, Montclair, NJ; ¹⁸Allina Health Minneapolis Heart Institute, Abbott Northwestern Hospital and Center for Coronary Artery Disease, Minneapolis Heart Institute Foundation Minneapolis; ¹⁹Baylor Scott and White the Heart Hospital–Plano, Plano, TX; ²⁰Ichinomiyanishi Hospital, Aichi, Japan; ²¹NYU Langone Health and NYU School of Medicine, Department of Medicine, Division of Cardiology, New York; ²²Lausanne University Center Hospital, Lausanne, Switzerland; ²³Department of Population Health Sciences, Weill Cornell Medicine, New York; ²⁴Department of Molecular and Clinical Medicine, Gothenburg University, Gothenburg, Sweden; ²⁵Department of Cardiology, Sahlgrenska University Hospital, Gothenburg, Sweden; ²⁶Tel Aviv University, Tel Aviv, Israel.

REFERENCES

1. Tonino PAL, De Bruyne B, Pijls NHJ, et al. Fractional flow reserve versus angiography for guiding percutaneous coronary intervention. *N Engl J Med* 2009;360:213-24.
2. De Bruyne B, Fearon WF, Pijls NHJ, et al. Fractional flow reserve-guided PCI for stable coronary artery disease. *N Engl J Med* 2014;371:1208-17.
3. Smits PC, Abdel-Wahab M, Neumann F-J, et al. Fractional flow reserve-guided multivessel angioplasty in myocardial infarction. *N Engl J Med* 2017;376:1234-44.
4. Davies JE, Sen S, Dehbi H-M, et al. Use of the instantaneous wave-free ratio or fractional flow reserve in PCI. *N Engl J Med* 2017;376:1824-34.
5. Götberg M, Christiansen EH, Gudmundsdottir IJ, et al. Instantaneous wave-free ratio versus fractional flow reserve to guide PCI. *N Engl J Med* 2017;376:1813-23.
6. Virani SS, Newby LK, Arnold SV, et al. 2023 AHA/ACC/ACCP/ASPC/NLA/PCNA guideline for the management of patients with chronic coronary disease: a report of the American Heart Association/American College of Cardiology Joint Committee on clinical practice guidelines. *Circulation* 2023;148(9):e9-e119.
7. Vrints C, Andreotti F, Koskinas KC, et al. 2024 ESC guidelines for the management of chronic coronary syndromes. *Eur Heart J* 2024;45:3415-537.
8. Parikh RV, Liu G, Plomondon ME, et al. Utilization and outcomes of measuring fractional flow reserve in patients with stable ischemic heart disease. *J Am Coll Cardiol* 2020;75:409-19.
9. Pellicano M, Lavi I, De Bruyne B, et al. Validation study of image-based fractional flow reserve during coronary angiography. *Circ Cardiovasc Interv* 2017;10(9):e005259.
10. Fearon WF, Achenbach S, Engström T, et al. Accuracy of fractional flow reserve derived from coronary angiography. *Circulation* 2019;139:477-84.
11. Witberg G, De Bruyne B, Fearon WF, et al. Diagnostic performance of angiogram-derived fractional flow reserve: a pooled analysis of 5 prospective cohort studies. *JACC Cardiovasc Interv* 2020;13:488-97.
12. Redfors B, Madhavan MV, Kirtane AJ, et al. FFRangio-guided versus pressure wire-guided PCI: design and rationale of the multicentre, randomised ALL-RISE trial. *EuroIntervention* 2025;21:961-9.
13. Andersen BK, Sejr-Hansen M, Maillard L, et al. Quantitative flow ratio versus fractional flow reserve for coronary revascularisation guidance (FAVOR III Europe): a multicentre, randomised, non-inferiority trial. *Lancet* 2024;404:1835-46.
14. Kang D-Y, Ahn J-M, Lee CH, et al. Deferred vs. performed revascularization for coronary stenosis with grey-zone fractional flow reserve values: data from the IRIS-FFR registry. *Eur Heart J* 2018;39:1610-9.
15. Witberg G, Bental T, Levi A, et al. Clinical outcomes of FFRangio-guided treatment for coronary artery disease. *JACC Cardiovasc Interv* 2022;15:468-70.
16. Shlofmitz E, Shin D, Alasnag M, et al. Angiography-derived physiology for coronary artery disease assessment: Expert opinion from a SCAI roundtable. *J Soc Cardiovasc Angiogr Interv* 2026 ([https://www.jsc.ai.org/article/S2772-9303\(25\)01602-3/fulltext](https://www.jsc.ai.org/article/S2772-9303(25)01602-3/fulltext)).
17. Andersen BK, Holm NR, Mogensen LJH, et al. Coronary revascularisation deferral based on quantitative flow ratio or fractional flow reserve: a post hoc analysis of the FAVOR III Europe trial. *EuroIntervention* 2025;21(3):e161-e170.
18. Kristensen SK, Holm MB, Maillard L, et al. Repeatability and quality assessment of QFR in the FAVOR III Europe trial: the REPEAT-QFR study. *EuroIntervention* 2026;22(1):e53-e65.
19. Daemen J, van der Eijk JA, Barbierato M, et al. Angiography-based physiology to guide coronary revascularization. *N Engl J Med* 2026;395:20-31.
20. Erriquez A, Colaiori I, Hakeem A, et al. Functional coronary angiography to indicate and guide revascularization in STEMI patients with multivessel disease: rationale and design of the AIR-STEMI trial. *Am Heart J* 2025;284:71-80.

Copyright © 2026 Massachusetts Medical Society.